Ordinary Polynomial Multiplication over Every Fixed Finite Field

A strict logarithmic saving on fixed tapes

Árpád Csizmadia
arpad.csizmadia.public@gmail.com

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Abstract

For each fixed finite field $\mathbb{F}_q$, ordinary polynomial multiplication has deterministic bit complexity $O_q(n(\log(n+2))^{1-2^{-38}})$ on a fixed finite-tape machine. For each fixed odd field $\mathbb{F}_{p^e}$, the stronger bound $O_{p,e}(n(\log(n+2))^{9999/10000})$ follows from the cyclotomic-host reduction proved here. The characteristic-two branch and its fixed-extension corollary are supplied by the accompanying controlled-tensor preprint [4]. The all-field bound uses the larger of the two exponents and does not assert the sharper odd-field exponent in characteristic two. Lean formalization remains unfinished; full formalization is planned for a later update.

AI assistance and formal verification. This work was developed with substantial generative AI assistance in mathematical exploration, algorithm and proof development, writing, coding, and literature review. OpenAI Codex [16] assisted release preparation and source inspection. Historical model versions were not consistently recorded; no complete model-level attribution is claimed. Full Lean formalization is unfinished and planned for a later update. Partial Lean work in the broader project is not included here and does not verify these complete theorems. Finite computational checks cover only their stated scopes. This manuscript is a working preprint open for mathematical review.

1 Prior work and source lineage

For each fixed prime p, Harvey, van der Hoeven and Lecerf obtained $O_p(n \log n\, 8^{\log^* n})$ bit operations [5]. Harvey and van der Hoeven improved this to the unconditional $O_p(n \log n\, 4^{\log^* n})$ benchmark on deterministic fixed tapes and supplied the cyclotomic hosts used here [1]. Fixed-degree basis reduction extends that comparison to every individually fixed finite field. Their later uniform finite-field $O(n \log q \log(n \log q))$ theorem assumes a Linnik constant $L < 1+2^{-1162}$ [6, Theorem 1.2]. Harvey's October 2026 discussion identifies the fixed-field problem as a separate issue from the new integer-multiplication claims [7, Section 4]; it is context, not review of this work.

These developments established the algebraic and bit-complexity framework on which this paper builds. The $n \log n$ scale is a natural next benchmark for the unconditional fixed-field problem: the earlier uniform theorem reaches that scale with its stated number-theoretic assumption, while the unconditional cyclotomic algorithm retains the iterated-logarithmic factor. Our claims fix the field and address that remaining gap through the inherited hosts, new transfer accounting and accompanying binary constructions.

The source lineage is explicit. OpenAI's finite-network and tape interchange method [8] is an antecedent. The selected odd transfer imports the precise stage-two machine theorem [2], not an end-to-end integer multiplier. The native supplement retains community chart and contributor attributions [9]; related mixed-recursion and exact-complex RAM work is credited separately [10]. General fixed-matrix tensor acceleration in Alman's lookup-table model [11] is prior art, with a different access model. The supplement uses classical integer arithmetic [12, 13] and classical linear/quadratic Clifford descriptions [14]; its specific community-derived coordinate constructions retain their own credit. These classical and community developments provide essential antecedents to the construction.

The contribution recorded here is the odd-field host transfer and its combination with the accompanying binary construction, under the stated field and machine quantifiers. The companion baseline note [15] instead combines the accessible binary route with the odd proposition below to obtain $O_q(n \log(n+2))$. These are related results with shared foundations.

2 Model and scope

The input consists of the ordinary coefficient lists of two polynomials of degree less than n over a fixed field $\mathbb{F}_q$, in a fixed polynomial basis. The output is the full ordinary coefficient list of their product. Time counts bit operations, including encoding, setup, movement, decoding, and output, on a deterministic machine with a fixed finite alphabet and a fixed finite number of sequential tapes. Constants may depend on the fixed field. No growing-field uniformity, practical cryptographic speedup, or combined workspace bound is claimed.

The logarithm is base two; changing its base changes only constants. All statements concern ordinary univariate multiplication. Group convolution, transformed products, integer multiplication, and multivariate products are different problems and are not used as output substitutes.

3 The odd-characteristic theorem

Proposition 1 (Odd-field ordinary product)**.** *For each fixed odd prime p and positive integer e, ordinary degree-$< n$ polynomial multiplication over $\mathbb{F}_{p^e}$ is deterministic and takes*

$$O_{p,e}\Big(n(\log(n+2))^{9999/10000}\Big)$$

bit operations on a fixed finite-tape machine.

Proof source and dependency closure. The detailed source proof is bundled at `proofs/odd_grouped_native_transfer.tex` and `proofs/odd_cyclotomic_host_transfer.tex`; exact source hashes and copy edits are recorded in `audit.json`. The complete current native construction, with the cited proof labels, is preserved in `proofs/native_odd_source_snapshot.tex.txt`. A dependency map and the remaining external import are documented in `proofs/PROOF_SOURCES.md`. The proposition is not inferred from an audit label or from a finite experiment.

The host is the unconditional fixed-prime cyclotomic construction of Harvey and van der Hoeven [1]: it supplies $2n < N < 4n$, admissible host primes, a divisor, and a principal root in $N^{o(1)}$ setup time. The coefficient host is enlarged by appending $O(\log\log N)$ unused host primes only after the public generator degrees are frozen. With $L_0 = \lceil \log_2 N \rceil$, $a = \lceil \log_2 L_0 \rceil$, and $R_0 = 2^{a^2}$, the enlargement yields

$$\log \alpha' = O(a^4), \qquad m'D' = \Theta(N), \qquad 1 - D'/\alpha' = O(a^2/L_0^3), \quad m' = N/\alpha', \quad D' = \varphi(\alpha').$$

The root lift $Y_{\mathrm{old}} \mapsto U^{\alpha'/\alpha}$ transports the old root identities; new input coefficients instead use $Y_{\mathrm{new}} \mapsto U$. This distinction preserves the ordinary coefficient interpretation. Both coefficient CRT components are reconstructed, the affine top map is inverted, and $2n-1 < N$ rules out cyclic aliasing.

The odd-characteristic formal compiler works over every commutative $\mathbb{F}_p$-algebra, including nonreduced algebras. It chooses radix three for $p \neq 3$ and radix five for $p = 3$, proves the root differences are units by polynomial Bezout identities, and explicitly normalizes the inverse by t^{-1}. Its recurrence gives a uniform quasilinear-size $O_p(v \log^{16}(2v))$ circuit of depth $O_p(\log^2(2v))$ after counted copy trees. The arbitrary-dirty replay uses a subtractive second read to cancel FUz; it does not assume clean scratch. Six fixed-kernel convolutions implement each exact-width Toeplitz chunk, with source, target, and gate ports all inside the parent volume. The resulting native schedule has $O_p(\log^{35}(2R))$ layers.

For the Newton factorization the sign $\beta_j = (-1)^j u^{j(j-1)/2}$ is retained. On the growing coefficient algebra, split-component unit selectors express each required shear as two unit shears. A fixed five-call word supplies the unit shear over a fixed quadratic field; its diagonal conjugations are explicitly priced as growing-ring operations. No fixed-kernel theorem is applied to the growing algebra. CRT characters are normalized by both inverse roots and $(m')^{-1}$, and the complement component is charged separately. Classical packed arithmetic is used independently for small coefficient-ring products, so there is no recursive call to the target multiplier.

The native supplier is the retained odd-ring phase/matching/CRT construction bundled in `proofs/native_odd_source_snapshot.tex.txt` (including the native dock, phase, selector, controlled operation, CRT, and stage-two phase results), with the native parameter proof used at the broader shape derived in the host transfer. The named non-elementary movement dependency is `IntegerMultBounds/MathlibTM2/M8/Stage2Machine/Bound.lean`, declaration `IntegerMultBounds.MathlibTM2.M8.Stage2Machine.Bound.uniform_bound` at commit `fefd699f9389af9610469977962613333539e5909` [2]; its precise source is linked in the bundled `proofs/PINNED_STAGE2_BINARY_MOVEMENT_BRIDGE_20261030.txt`. The bridge derives the arbitrary-width adapter and records the exact source hash. This is a named prior theorem, not a new unproved hypothesis. The final cost is

$$O_p\Big(NL_0^{\gamma_0}(\log L_0)^C\Big) + N \operatorname{polylog} L_0 + N^{o(1)}, \qquad \gamma_0 = 1 - \frac{619}{5000000} < \frac{9999}{10000};$$

the strict exponent gap absorbs all fixed secondary logarithms. Fixed extensions use a constant number of prime-field products and fixed basis maps. The complete itemized audit is `proofs/ODD_CYCLOTOMIC_FULL_ADVERSARIAL_AUDIT_20261030.txt`. Its finite C++ witness checks bounded identities and examples only; the asymptotic conclusion follows from the derivation and cited interfaces, not from those samples. □

Lemma 1 (Fixed binary-basis extension reduction)**.** *If ordinary multiplication over $\mathbb{F}_2$ on degree-$< n$ inputs has bit cost $T(n)$, then for each fixed $s \geq 1$, ordinary multiplication over $\mathbb{F}_{2^s}$ in a fixed polynomial basis has bit cost $O_s(T(n)+n)$ on the same fixed-tape model.*

Proof. Fix an $\mathbb{F}_2$-basis $\omega_1, \ldots, \omega_s$ of $\mathbb{F}_{2^s}$. Write the input coefficient sequences as $f = \sum_{i=1}^{s} \omega_i f_i$ and $g = \sum_{j=1}^{s} \omega_j g_j$, where $f_i, g_j \in \mathbb{F}_2[X]$ have degree less than n. Compute the s^2 ordinary binary products $f_i g_j$. For each basis coordinate k, combine their coefficients using the fixed structure constants $\omega_i \omega_j = \sum_k c_{ijk} \omega_k$. The fixed basis encoding, the $O(s^3 n)$ additions/scalar maps, and the final coefficient decoding cost $O_s(n)$. Since s is fixed, the s^2 binary calls cost $O_s(T(n)+n)$, proving the claim. □

Theorem 1 (All-fixed-field bound)**.** *For every fixed finite field $\mathbb{F}_q$, ordinary degree-$< n$ multiplication in a fixed polynomial basis takes*

$$O_q\Big(n(\log(n+2))^{1-2^{-38}}\Big)$$

deterministic bit operations on a fixed finite-tape machine.

Proof. Write $q = p^e$. If $p = 2$, the controlled-tensor theorem and its fixed extension corollary in [4] give the stated bound. If p is odd, Proposition 1 gives the stronger exponent $9999/10000$. Since $9999/10000 < 1 - 2^{-38}$, its logarithmic factor is bounded by the one displayed here for $n \geq 2$; finitely many smaller lengths are absorbed in the field-dependent constant. Both branches use ordinary coefficient representations and the fixed finite-tape bit-time model. All field and basis constants are independent of n. □

Theorem 2 (Unconditional fixed-field baseline)**.** *Every fixed finite field has an unconditional ordinary multiplication algorithm in $O_q(n \log(n+2))$ bit operations on the stated machine.*

Proof. This follows immediately from Theorem 1, because $(\log(n+2))^{1-2^{-38}} = O(\log(n+2))$. The independent Pascal-transform route for characteristic two is also recorded in [3]. The odd fields follow from Proposition 1. □

4 Verification and limitations

The bundled historical 26-item adversarial derivation audit records a bounded whole-host witness (24 products and 1,920 assertions over $\mathbb{F}_3$ and $\mathbb{F}_5$); its implementation is not included here. The recorded witness checks finite map boundaries; it does not establish the growing tape algorithms or asymptotic bound. The characteristic-two claim is imported from the accompanying controlled-tensor preprint, whose result is stated with its own proof scope. The general bound combines that result with the odd proposition.

Neither the odd proposition nor the combined theorem has an end-to-end Lean proof in this package. Full Lean formalization is unfinished and planned for a later update. The package records no independent human review, publication acceptance, novelty certification, or practical speedup.

References

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[3] Árpád Csizmadia. *Exact Binary Polynomial Multiplication from Fast Pascal Transforms.* Accompanying preprint in this collection; source: `papers/02-simple-binary/main.tex`.

[4] Árpád Csizmadia. *Ordinary Binary Polynomial Multiplication below $n \log n$.* Subtitle: Controlled tensor routing on fixed tapes. Accompanying preprint in this collection; source: `papers/03-best-binary/main.tex`.

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